

William Learned A1 Cloud Computing Final Task

Brief Summary

Before getting started, I had to mentally prepare. I finished the previous assignment weeks ago, so it had been a while since I worked with AWS. I decided to get the database up and running first. Then, I got the API up and running. Finally, I uploaded the modified build to a new bucket.

Instead of nano, I used micro as a text editor, which was a much better experience--micro can actually scroll. I used forever to run the API index file as a daemon, but I think I like pm2 better. It was temporary, so it didn't matter.

Everything went very well with very few issues. It was annoying to not have any SSL certificates, so I kept seeing insecure warnings. I considered fixing that, but the machines and bucket would be deleted soon. Plus, without a qualified domain name, I thought it would be silly. However, I do want to practice securing websites with certificates for my own knowledge and experience.

After executing the commands documented below, I created a new s3 bucket and uploaded a modified build folder that referenced this new API address. Then, I set the bucket properties for static web hosting and set the bucket policy to allow public visitors. I visited the site in the browser and took a screenshot after signing in as a user.

Then, everything was DELETED FOREVER :)

AWS & Linux Commands

This is the machine for the database:

```
$ aws ec2 run-instances --image-id ami-0a0e5d9c7acc336f1 --count 1 --instance-type t2.micro --key-name wll-key-pair --security-group-ids sg-0ecc23566b975ad0b --subnet-id subnet-03008a96b10546d14
```

This is the machine for the API:

```
$ aws ec2 run-instances --image-id ami-0a0e5d9c7acc336f1 --count 1 --instance-type t2.micro --key-name wll-key-pair --security-group-ids sg-0ecc23566b975ad0b --subnet-id subnet-03008a96b10546d14
```

Connecting to the database instance:

```
$ ssh -i wll-key-pair.pem ubuntu@3.239.184.255 $ sudo apt update $ sudo apt install software-properties-common gnupg apt-transport-https ca-certificates -y $ curl -fsSL https://www.mongodb.org/static/pgp/server-7.0.asc | sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg --dearmor
```

```
$ echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 multiverse" | sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list $ sudo apt update $ sudo install mongodb-community
```

```
$ sudo apt install micro $ micro users.json $ micro movies.json
```

```
$ sudo systemctl start mongod $ sudo systemctl enable mongod $ systemctl status mongod
```

```
$ cat users.json $ mongoimport --db=cineDataDB --collection=users --file=users.json $ mongoimport --db=cineDataDB --collection=movies --file=movies.json
```

```
$ mongosh $ use cineDataDB $ db.users.find() $ exit
```

```
$ sudo micro /etc/mongod.conf $ bindIp: 0.0.0.0 $ sudo systemctl restart mongod $ aws ec2 authorize-security-group-ingress --group-id sg-0ecc23566b975ad0b --protocol tcp --port 27017 --cidr 172.31.0.0/16
```

Now on the API machine:

```
$ ssh -i wll-key-pair.pem ubuntu@44.200.196.201 $ sudo apt update $ sudo apt install micro git nginx curl -y $ sudo apt install npm nodejs # I forgot that the version of node apt can install is very old $ node -v $ curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.3/install.sh | bash #fnm is another nice option to manage node $ export NVM_DIR="$HOME/.nvm" [ -s "$NVM_DIR/nvm.sh" ] && . "$NVM_DIR/nvm.sh" # This loads nvm [ -s "$NVM_DIR/bash_completion" ] && . "$NVM_DIR/bash_completion" # This loads nvm bash_completion \ source ~/.bashrc $ nvm install --lts $ git clone https://github.com/animalelder/movie_api.git $ npm i $ npm audit fix
```

```
$ sudo micro /etc/nginx/sites-available/app $ sudo ln -s /etc/nginx/sites-available/app /etc/nginx/sites-enabled/ $ sudo rm /etc/nginx/sites-enabled/default $ sudo systemctl start nginx $ sudo systemctl enable nginx
```

```
$ micro index.js $ npm install -g forever $ npm run start $ nvm install 22.9 && nvm use 22.9
```

```
$ micro .env #adding connection URI and port to environment variables $ forever index.js
```

Screenshots

EC2

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:v=3;\$case=tags:true%5c,client:false;\$regex=tags:false%5c,client:fal...

Services

Search

[Option+S]

N. Virginia

Liam @ 3111-4154-6316

Lambda

EC2

S3

IAM

VPC

EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity

Reservations

Images

AMIs

AMI Catalog

Instances (2)

Info

Last updated less than a minute ago

Refresh

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

1

Table with 7 columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm. It contains 2 rows of instance data.

S3

Amazon S3 > Buckets > cinedata

cinedata

Info

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (21)

Info

Refresh

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

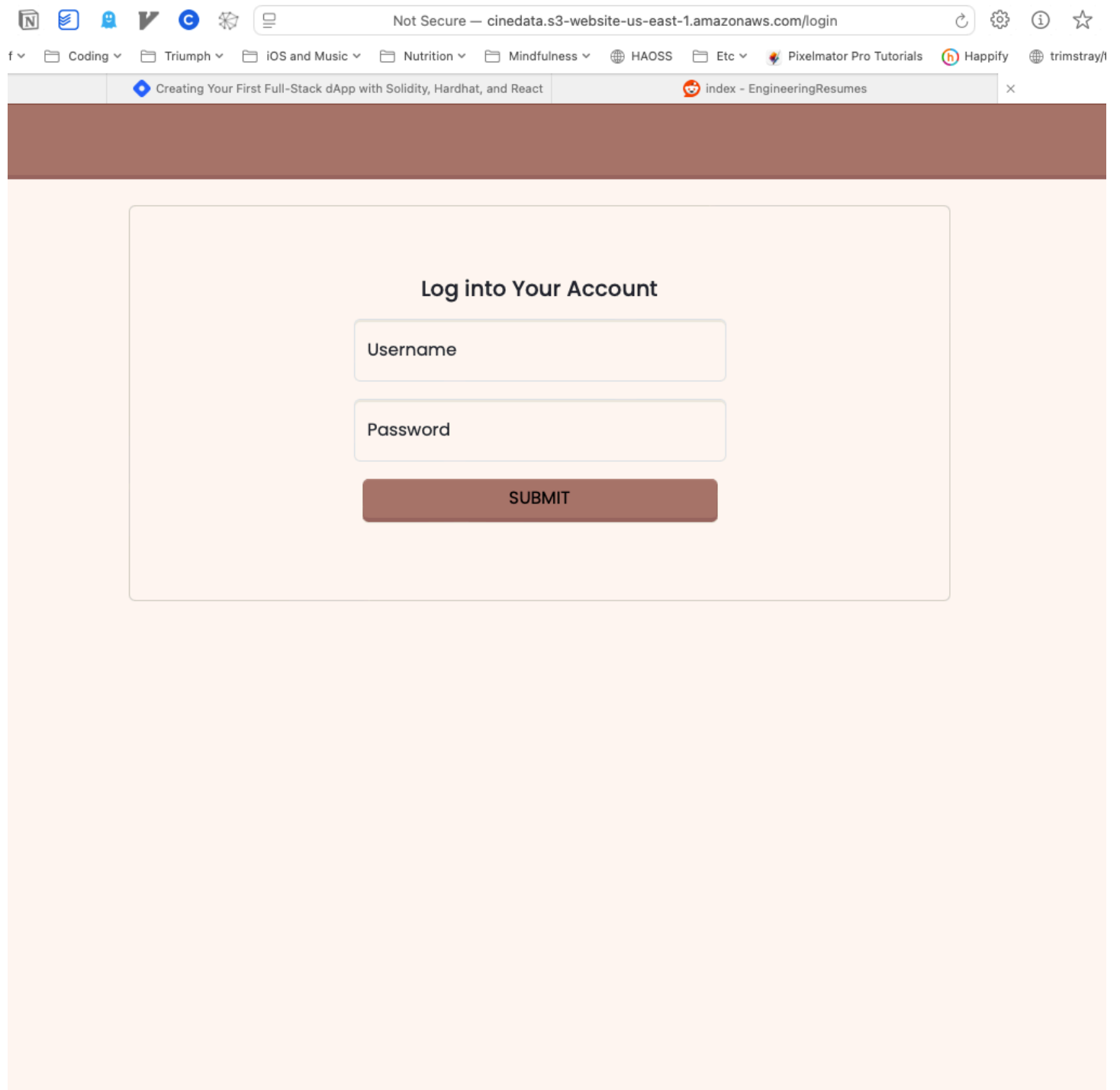
Find objects by prefix

< 1 > ⚙

☐	Name ▲	Type ▼	Last modified ▼	Size ▼	Storage class ▼
☐	 index.2212206d.css	css	September 23, 2024, 15:21:56 (UTC-07:00)	275.0 KB	Standard
☐	 index.2212206d.css.map	map	September 23, 2024, 15:21:56 (UTC-07:00)	29.4 KB	Standard
☐	 index.ae8edde4.css	css	September 23, 2024, 15:21:57 (UTC-07:00)	224.4 KB	Standard
☐	 index.ae8edde4.css.map	map	September 23, 2024, 15:21:58 (UTC-07:00)	503.5 KB	Standard
☐	 index.b1bccf1c.css	css	September 23, 2024, 15:21:58 (UTC-07:00)	224.5 KB	Standard
☐	 index.b1bccf1c.css.map	map	September 23, 2024, 15:21:59 (UTC-07:00)	505.4 KB	Standard
☐	 index.b4b6dfad.js	js	September 23, 2024, 15:22:00 (UTC-07:00)	3.5 MB	Standard
☐	 index.b4b6dfad.js.map	map	September 23, 2024, 15:22:01 (UTC-07:00)	3.7 MB	Standard

September 23, 2024

LOGIN



Not Secure — cinedata.s3-website-us-east-1.amazonaws.com/login

Creating Your First Full-Stack dApp with Solidity, Hardhat, and React

index - EngineeringResumes

Log into Your Account

Username

Password

SUBMIT

Authenticated index

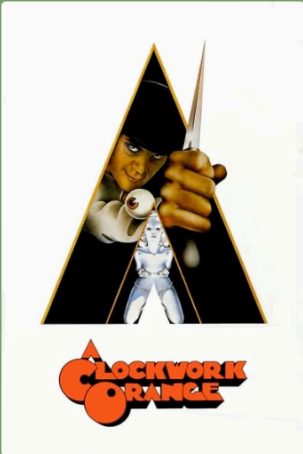
Not Secure — cinedata.s3-website-us-east-1.amazonaws.com

ScrapeMasterCreating Your First Full-Stack dApp with Solidity, Ha...Index - EngineeringResumesx

cineData

See All of the MoviesProfileLogout

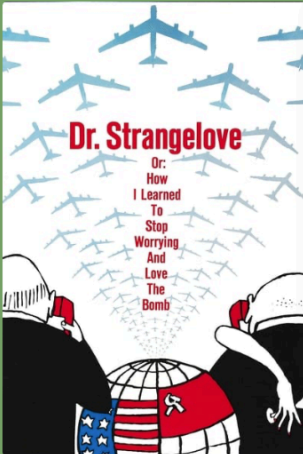
Search the films on cineData...



A Clockwork Orange

DRAMA

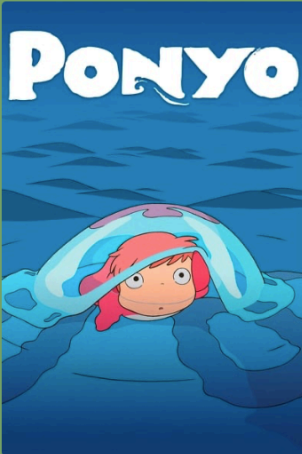
♡ MORE



Dr. Strangelove

COMEDY

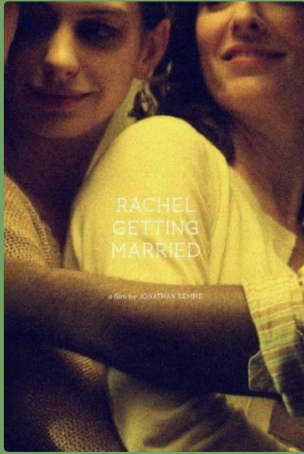
♡ MORE



Ponyo

ANIMATED

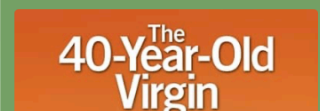
♡ MORE



Rachel Getting Married

DRAMA

♡ MORE



The 40-Year-Old Virgin



PRINCESS MONONOKE





SPIRITED AWAY